

Qatar-1b

R-1551

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_220412 · created 2026-08-01T20:00:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459681.9264 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.11 +/- 0.033
Transit depth (Rp/Rs)^2	1.21 +/- 0.73 [%]
Orbital Inclination (inc)	88.95 +/- 2.62
Airmass coefficient 1 (a1)	1.251 +/- 0.026
Airmass coefficient 2 (a2)	-0.16 +/- 0.014
Scatter in the residuals of the lightcurve fit is	2.11 %
Best Comparison Star	None
Optimal Aperture	3.39
Optimal Annulus	11.31
Transit Duration (day)	0.0773 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.11 ± 0.033 vs 0.1463 (deviation 1.03σ)
Duration fit vs published	0.0773 d vs 0.0692 d (deviation 1.31σ)
Mid-time O–C	5.7 min
Depth SNR	3.1
Residual scatter	2.11 %

Light curve

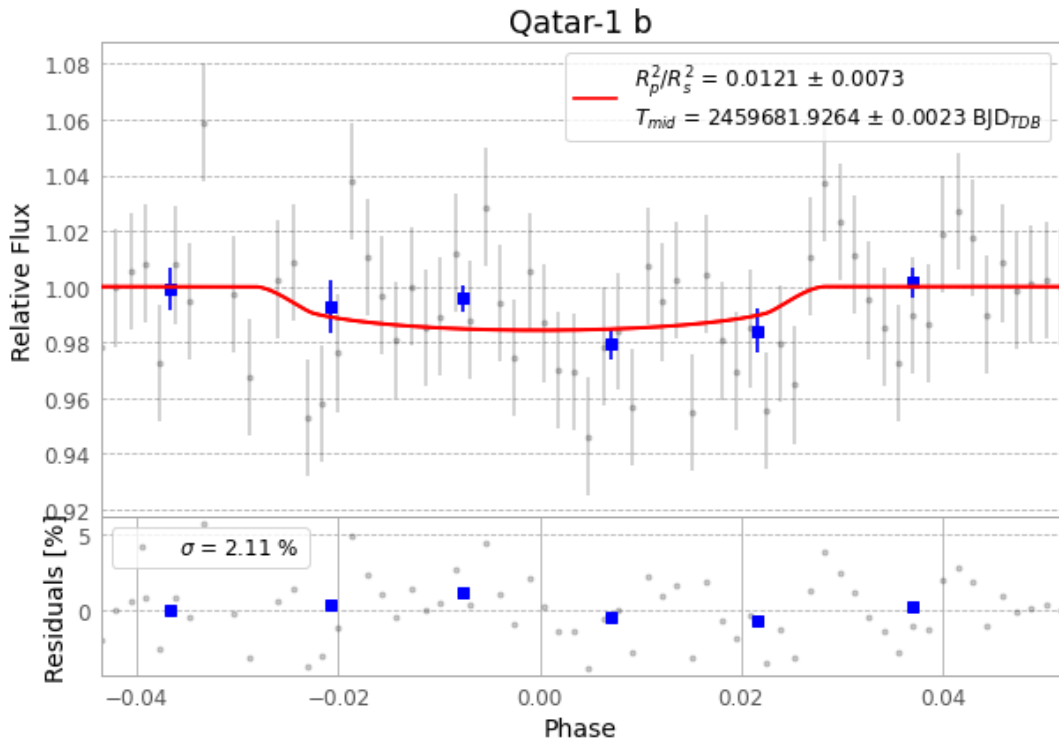


Figure 1 — FinalLightCurve_Qatar-1 b_2022-04-12.png (SHA-256 1cd1e8ac7aaaa680...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [260, 363], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d87126f4c19773be...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), Qatar-1220412084512.FITS → Qatar-1220412120012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-220412.log (f3c45d8c8345...), FinalLightCurve_Qatar-1 b_2022-04-12.png (1cd1e8ac7aaa...), FinalLightCurve_Qatar-1 b_2022-04-12.csv (7532544f929d...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 20:00Z.